The average velocity is defined as ratio of the displacement vector to the corresponding time interval

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t}.$$

It is a vector quantity. The direction of average velocity is in the direction of the displacement vector ($\Delta \vec{r}$).

This is also shown in Figure 2.34.

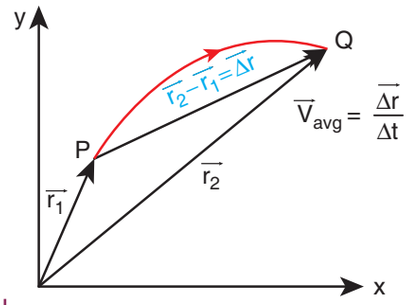


Figure 2.34 Average velocity

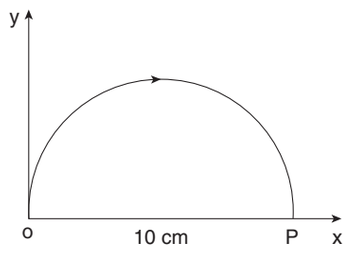
Average speed

The average speed is defined as the ratio of total path length travelled by the particle in a time interval.

Average speed = total path length / total time

EXAMPLE 2.20

Consider an object travelling in a semi-circular path from point O to point P in 5 second, as shown in the Figure given below. Calculate the average velocity and average speed.



Solution

Average velocity $\vec{v}_{avg} = \frac{\vec{r}_P - \vec{r}_O}{\Delta t}$

Here $\Delta t = 5\ s$

$\vec{r}_O = 0, \vec{r}_P = 10\hat{i}$

$$\vec{v}_{avg} = \frac{10\hat{i}\ cm}{5\ s} = 2\hat{i}\ cm\ s^{-1}.$$

The average velocity is in the positive x direction.

The average speed = total path length / time taken (the path is semi-circular)

$$= \frac{5\pi\ cm}{5\ s} = \pi\ cm\ s^{-1} \approx 3.14\ cm\ s^{-1}$$

Note that the average speed is greater than the magnitude of the average velocity.

Instantaneous velocity or velocity

The instantaneous velocity at an instant t or simply 'velocity' at an instant t is defined as limiting value of the average velocity as $\Delta t \to 0$, evaluated at time t.

In other words, velocity is equal to rate of change of position vector with respect to time. Velocity is a vector quantity.

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

In component form, this velocity is

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}\left(x\hat{i} + y\hat{j} + z\hat{k}\right)$$
$$= \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}.$$

Here $\frac{dx}{dt} = v_x = x-$ component of velocity

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